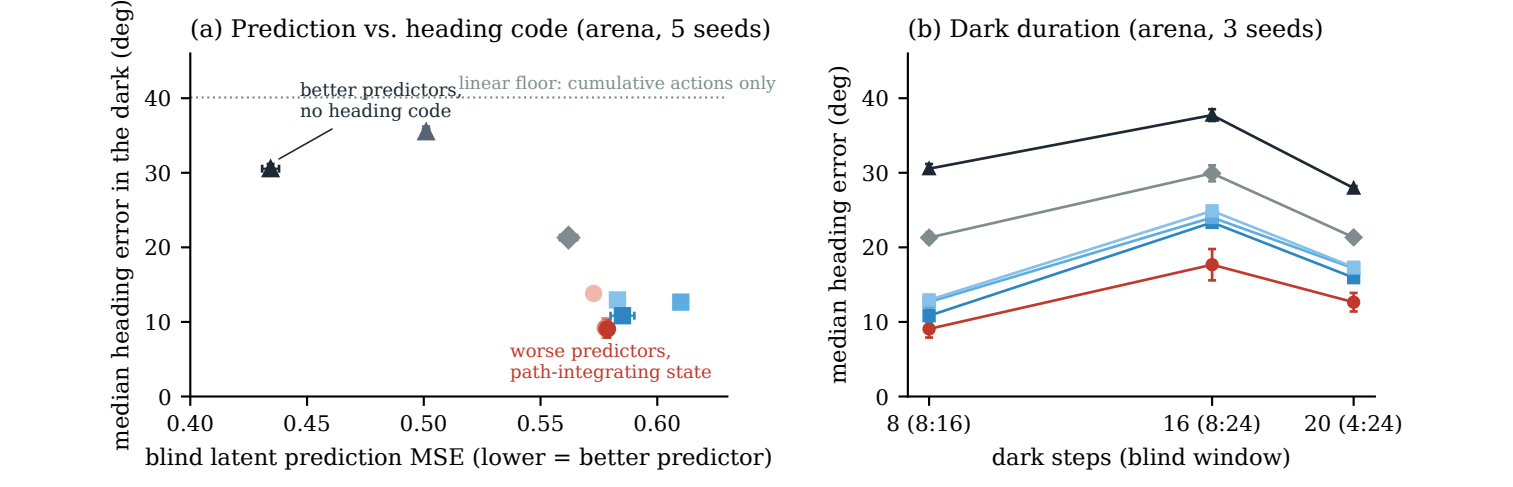
(a) Prediction vs. heading code (arena, 5 seeds)
median heading error in the dark (deg)
better predictors,
no heading code
linear floor: cumulative actions only
worse predictors,
path-integrating state
0
10
20
30
40
0.40
0.45
0.50
0.55
0.60
blind latent prediction MSE (lower = better predictor)
(b) Dark duration (arena, 3 seeds)
median heading error (deg)
8 (8:16)
16 (8:24)
20 (4:24)
dark steps (blind window)
GRU, parameter-matched
GRU, recurrent-budget-matched
dense
connectome, unsigned
connectome, random signs
connectome
random
block-shuffled
degree-shuffled